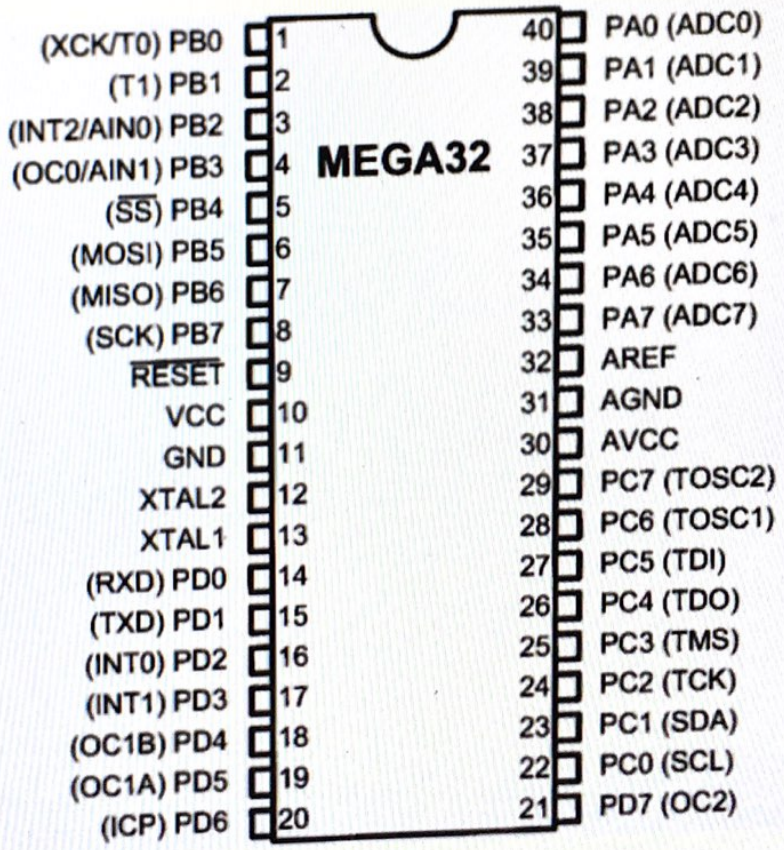
Chapter 4-Fig4-1



Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	RW	RW	RW	RW	RW	RW	RW
Initial Value	0	0	0	0	0	0	0	0

FOC0 D7 Force compare match: This is a write-only bit, which can be used while generating a wave. Writing 1 to it causes the wave generator to act as if a compare match had occurred.

WGM00, WGM01

D6	D3	Timer0 mode selector bits
0	0	Normal
0	1	CTC (Clear Timer on Compare Match)
1	0	PWM, phase correct
1	1	Fast PWM

COM01:00 D5 D4 Compare Output Mode:
These bits control the waveform generator (see Chapter 15).

CS02:00

D2	D1	D0	Timer0 clock selector
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (No Prescaling)
0	1	0	clk / 8
0	1	1	clk / 64
1	0	0	clk / 256
1	0	1	clk / 1024
1	1	0	External clock source on T0 pin. Clock on falling edge.
1	1	1	External clock source on T0 pin. Clock on rising edge.

Figure 9-5. TCCR0 (Timer/Counter Control Register) Register

Bit	7	6	5	4	3	2	1	0
	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

TOV0 D0 Timer0 overflow flag bit
0 = Timer0 did not overflow.
1 = Timer0 has overflowed (going from $FF to $00).

OCF0 D1 Timer0 output compare flag bit
0 = compare match did not occur.
1 = compare match occurred.

TOV1 D2 Timer1 overflow flag bit

OCF1B D3 Timer1 output compare B match flag

OCF1A D4 Timer1 output compare A match flag

ICF1 D5 Input Capture flag

TOV2 D6 Timer2 overflow flag

OCF2 D7 Timer2 output compare match flag

Figure 9-6. TIFR (Timer/Counter Interrupt Flag Register)

D7							D0
OCIE2	TOIE2	TICIE1	OCIE1A	OCIE1B	TOIE1	OCIE0	TOIE0

TOIE0 Timer0 overflow interrupt enable
= 0 Disables Timer0 overflow interrupt
= 1 Enables Timer0 overflow interrupt

OCIE0 Timer0 output compare match interrupt enable
= 0 Disables Timer0 compare match interrupt
= 1 Enables Timer0 compare match interrupt

TOIE1 Timer1 overflow interrupt enable
= 0 Disables Timer1 overflow interrupt
= 1 Enables Timer1 overflow interrupt

OCIE1B Timer1 output compare B match interrupt enable
= 0 Disables Timer1 compare B match interrupt
= 1 Enables Timer1 compare B match interrupt

OCIE1A Timer1 output compare A match interrupt enable
= 0 Disables Timer1 compare A match interrupt
= 1 Enables Timer1 compare A match interrupt

TICIE1 Timer1 input capture interrupt enable
= 0 Disables Timer1 input capture interrupt
= 1 Enables Timer1 input capture interrupt

TOIE2 Timer2 overflow interrupt enable
= 0 Disables Timer2 overflow interrupt
= 1 Enables Timer2 overflow interrupt

OCIE2 Timer2 output compare match interrupt enable
= 0 Disables Timer2 compare match interrupt
= 1 Enables Timer2 compare match interrupt

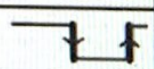
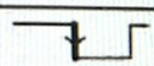
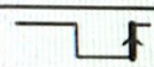
These bits, along with the I bit, must be set high for an interrupt to be responded to. Upon activation of the interrupt, the I bit is cleared by the AVR itself to make sure another interrupt cannot interrupt the microcontroller while it is servicing the current one. At the end of the ISR, the RETI instruction will make I = 1 to allow another interrupt to come in.

Figure 10-3. TIMSK (Timer Interrupt Mask) Register

D7 D0

SE	SM2	SM1	SM0	ISC11	ISC10	ISC01	ISC00

ISC01, ISC00 (Interrupt Sense Control bits) These bits define the level or edge on the external INT0 pin that activates the interrupt, as shown in the following table:

ISC01	ISC00		Description
0	0		The low level of INT0 generates an interrupt request.
0	1		Any logical change on INT0 generates an interrupt request.
1	0		The falling edge of INT0 generates an interrupt request.
1	1		The rising edge of INT0 generates an interrupt request.

ISC11, ISC10 These bits define the level or edge that activates the INT1 pin.

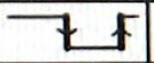
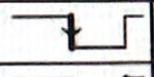
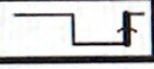
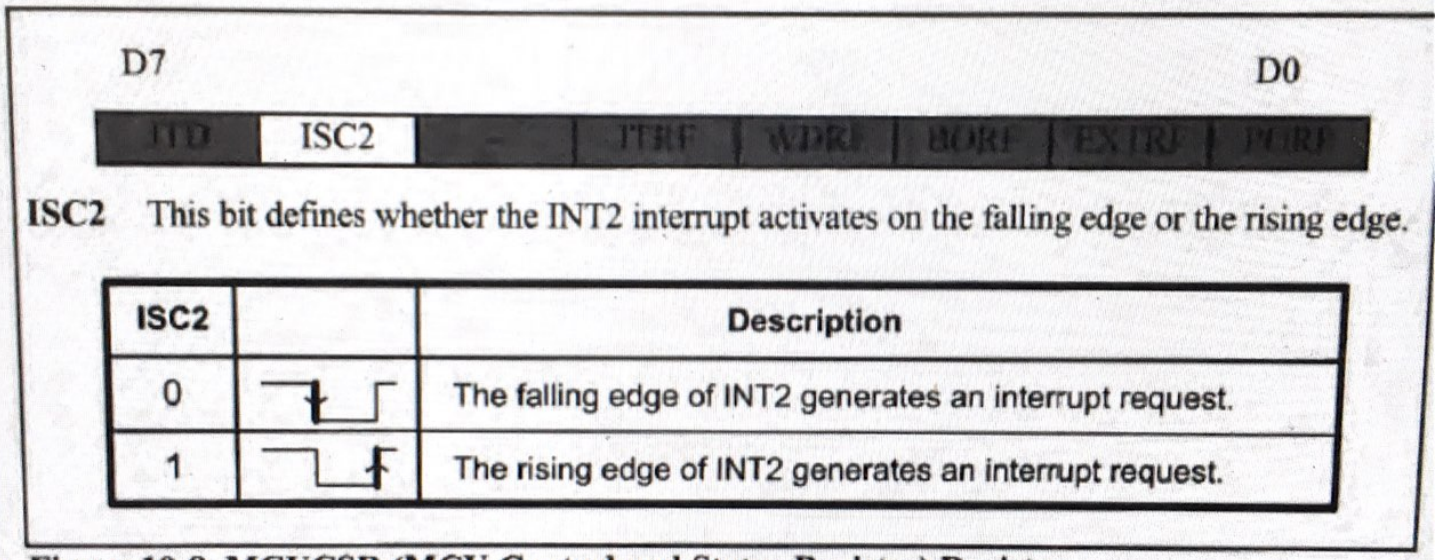
ISC11	ISC10		Description
0	0		The low level of INT1 generates an interrupt request.
0	1		Any logical change on INT1 generates an interrupt request.
1	0		The falling edge of INT1 generates an interrupt request.
1	1		The rising edge of INT1 generates an interrupt request.

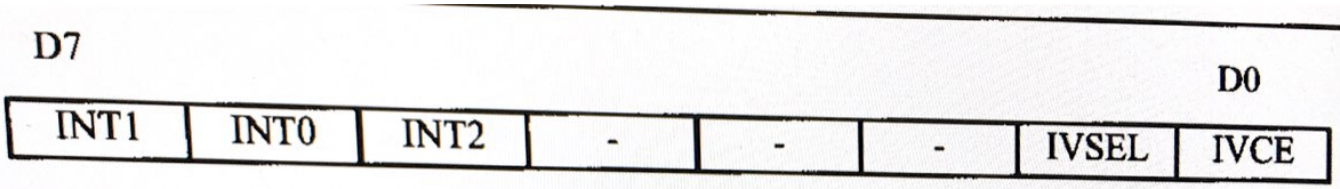
Figure 10-7. MCUCR (MCU Control Register) Register



ISC2 This bit defines whether the INT2 interrupt activates on the falling edge or the rising edge.

ISC2		Description
0		The falling edge of INT2 generates an interrupt request.
1		The rising edge of INT2 generates an interrupt request.

Figure 10-8. MCUCSR (MCU Control and Status Register) Register



INT0 External Interrupt Request 0 Enable
= 0 Disables external interrupt 0
= 1 Enables external interrupt 0

INT1 External Interrupt Request 1 Enable
= 0 Disables external interrupt 1
= 1 Enables external interrupt 1

INT2 External Interrupt Request 2 Enable
= 0 Disables external interrupt 2
= 1 Enables external interrupt 2

These bits, along with the I bit, must be set high for an interrupt to be responded to.

Figure 10-6. GICR (General Interrupt Control Register) Register

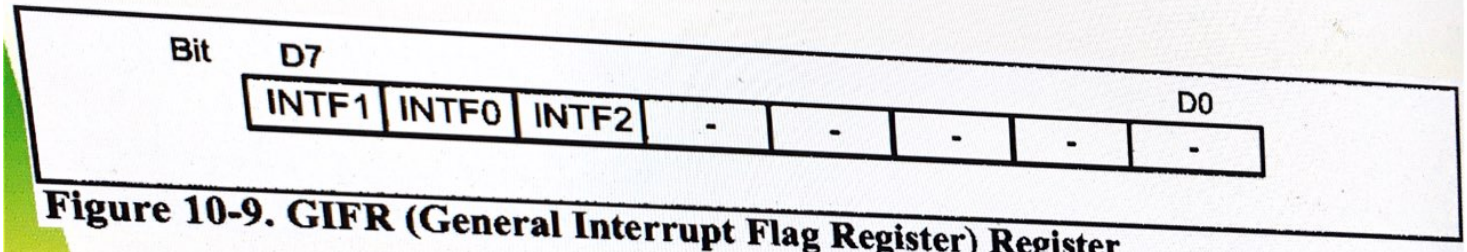


Figure 10-9. GIFR (General Interrupt Flag Register) Register

Atmega32 Interrupt Vector Table

Table 10-1: Interrupt Vector Table for the ATmega32 AVR

Interrupt	ROM Location (Hex)
Reset	0000
External Interrupt request 0	0002
External Interrupt request 1	0004
External Interrupt request 2	0006
Time/Counter2 Compare Match	0008
Time/Counter2 Overflow	000A
Time/Counter1 Capture Event	000C
Time/Counter1 Compare Match A	000E
Time/Counter1 Compare Match B	0010
Time/Counter1 Overflow	0012
Time/Counter0 Compare Match	0014
Time/Counter0 Overflow	0016
SPI Transfer complete	0018
USART, Receive complete	001A
USART, Data Register Empty	001C
USART, Transmit Complete	001E
ADC Conversion complete	0020
EEPROM ready	0022
Analog Comparator	0024
Two-wire Serial Interface (I2C)	0026
Store Program Memory Ready	0028

ATmega324pb Interrupt Vector Table

14.1 Interrupt Vectors in ATmega324PB

Table 14-1. Reset and Interrupt Vectors in ATmega324PB

Vector No	Program Address(2)	Source	Interrupts definition
1	0x0000(1)	RESET	External Pin, Power-on Reset, Brown-out Reset and Watchdog System Reset
2	0x0002	INT0	External Interrupt Request 0
3	0x0004	INT1	External Interrupt Request 1
4	0x0006	INT2	External Interrupt Request 2
5	0x0008	PCINT0	Pin Change Interrupt Request 0
6	0x000A	PCINT1	Pin Change Interrupt Request 1
7	0x000C	PCINT2	Pin Change Interrupt Request 2
8	0x000E	PCINT3	Pin Change Interrupt Request 3
9	0x0010	WDT	Watchdog Time-out Interrupt
10	0x0012	TIMER2_COMPA	Timer/Counter2 Compare Match A
11	0x0014	TIMER2_COMPB	Timer/Coutner2 Compare Match B
12	0x0016	TIMER2_OVF	Timer/Counter2 Overflow
13	0x0018	TIMER1_CAPT	Timer/Counter1 Capture Event
14	0x001A	TIMER1_COMPA	Timer/Counter1 Compare Match A
15	0x001C	TIMER1_COMPB	Timer/Coutner1 Compare Match B
16	0x001E	TIMER1_OVF	Timer/Counter1 Overflow
17	0x0020	TIMER0_COMPA	Timer/Counter0 Compare Match A
18	0x0022	TIMER0_COMPB	Timer/Coutner0 Compare Match B
19	0x0024	TIMER0_OVF	Timer/Counter0 Overflow
20	0x0026	SPI0_STC	SPI0 Serial Transfer Complete
21	0x0028	USART0_RX	USART0 Rx Complete
22	0x002A	USART0_UDRE	USART0, Data Register Empty

http://ww1.microchip.com/downloads/en/DeviceDoc/40001908A.pdf

Table 10-3: Interrupt Vector Name for the ATmega32/ATmega16 in WinAVR

Interrupt	Vector Name in WinAVR
External Interrupt request 0	INT0_vect
External Interrupt request 1	INT1_vect
External Interrupt request 2	INT2_vect
Time/Counter2 Compare Match	TIMER2_COMP_vect
Time/Counter2 Overflow	TIMER2_OVF_vect
Time/Counter1 Capture Event	TIMER1_CAPT_vect
Time/Counter1 Compare Match A	TIMER1_COMPA_vect
Time/Counter1 Compare Match B	TIMER1_COMPB_vect
Time/Counter1 Overflow	TIMER1_OVF_vect
Time/Counter0 Compare Match	TIMER0_COMP_vect
Time/Counter0 Overflow	TIMER0_OVF_vect
SPI Transfer complete	SPI_STC_vect
USART, Receive complete	USART0_RX_vect
USART, Data Register Empty	USART0_UDRE_vect
USART, Transmit Complete	USART0_TX_vect
ADC Conversion complete	ADC_vect
EEPROM ready	EE_RDY_vect
Analog Comparator	ANALOG_COMP_vect
Two-wire Serial Interface	TWI_vect
Store Program Memory Ready	SPM_RDY_vect

Address		Name
Mem.	I/O	
$20	$00	TWBR
$21	$01	TWSR
$22	$02	TWAR
$23	$03	TWDR
$24	$04	ADCL
$25	$05	ADCH
$26	$06	ADCSRA
$27	$07	ADMUX
$28	$08	ACSR
$29	$09	UBRRL
$2A	$0A	UCSRB
$2B	$0B	UCSRA
$2C	$0C	UDR
$2D	$0D	SPCR
$2E	$0E	SPSR
$2F	$0F	SPDR
$30	$10	PIND
$31	$11	DDRD
$32	$12	PORTD
$33	$13	PINC
$34	$14	DDRC
$35	$15	PORTC

Address		Name
Mem.	I/O	
$36	$16	PINB
$37	$17	DDRB
$38	$18	PORTB
$39	$19	PINA
$3A	$1A	DDRA
$3B	$1B	PORTA
$3C	$1C	EECR
$3D	$1D	EEDR
$3E	$1E	EEARL
$3F	$1F	EEARH
$40	$20	UBRRC
		UBRRH
$41	$21	WDTCR
$42	$22	ASSR
$43	$23	OCR2
$44	$24	TCNT2
$45	$25	TCCR2
$46	$26	ICR1L
$47	$27	ICR1H
$48	$28	OCR1BL
$49	$29	OCR1BH
$4A	$2A	OCR1AL

Address		Name
Mem.	I/O	
$4B	$2B	OCR1AH
$4C	$2C	TCNT1L
$4D	$2D	TCNT1H
$4E	$2E	TCCR1B
$4F	$2F	TCCR1A
$50	$30	SFIOR
$51	$31	OCDR
		OSCCAL
$52	$32	TCNT0
$53	$33	TCCR0
$54	$34	MCUCSR
$55	$35	MCUCR
$56	$36	TWCR
$57	$37	SPMCR
$58	$38	TIFR
$59	$39	TIMSK
$5A	$3A	GIFR
$5B	$3B	GICR
$5C	$3C	OCR0
$5D	$3D	SPL
$5E	$3E	SPH
$5F	$3F	SREG

Note: Although memory address $20-$5F is set aside for I /O registers (SFR) we can access them as I /O locations with addresses starting at $00.

Figure 2-7. I/O Registers of the ATmega32 and Their Data Memory Address Locations

Example

USER TIMER0 AND TIMER2 INTERRUPTS SIMULTANEOUSLY TO GENERATE 25 kHz and 50 kHz SQUARE WAVES ON PINS PB1 & PB7, RESPECTIVELY, WHILE DATA IS BEING TRANSFERRED FROM PORTC TO PORTD. USE fosc = 8 MHz.

TIMER0 (25 kHz) Use TIMER0 for both ATmega32

T0 = 1/25000 = 40 10^-6 s = 40 us
T0half = 20 us
TMC = (1 clock cycle/1 MC) (1s/8 10^6 clock cycle) = 0.125 us

20 us = n0half TMC
20 us = n0half (0.125 us)
n0half = 160 MCs for both ATmega32 and ATmega324pb

TIMER2 (50 kHz) (ATmega32)and ATMEGA324PB

T2 = 1/50000 = 20 10^-6 s = 20 us
T2half = 10 us
TMC = 0.125 us (same as above)

10 us = n2half TMC
10 us = n2half (0.125 us)
n2half = 80 MCs for ATmega32 and ATmega324pb

```
;
; TimerInterruptExAVR.asm

; ATMEGA32


; USER TIMER0 AND TIMER2 INTERRUPTS SIMULTANEOUSLY TO GENERATE SQUARE WAVES ON PINS PB1 & PB7, RESPECTIVELY WHILE
; DATA IS BEING TRANSFERRED FROM PORTC TO PORTD.

.ORG 0
JMP MAIN; JUMP OVER INTERRUPT VECTOR TABLE

; ----- ATMEGA32 INTERRUPT VECTOR TABLE
;.ORG 0x0A
;JMP T2_OV_ISR; TIMER 2 INTERRUPT SERVICE ROUTINE - JUMP TO PLACE IN CODE WITH ENOUGH SPACE TO IMPLEMENT ISR

;.ORG 0x16
;JMP T0_OV_ISR; TIMER 0 INTERRUPT SERVICE ROUTINE - JUMP TO ANOTHER CODE LOCATION WITH MORE SPACE


; ------ ATMEGA324PB INTERRUPT VECTOR TABLE
.ORG 0x16
JMP T2_OV_ISR; TIMER 2 INTERRUPT SERVICE ROUTINE - JUMP TO PLACE IN CODE WITH ENOUGH SPACE TO IMPLEMENT ISR

.ORG 0x24
JMP T0_OV_ISR; TIMER 0 INTERRUPT SERVICE ROUTINE - JUMP TO ANOTHER CODE LOCATION WITH MORE SPACE


.ORG 0x100
MAIN:   LDI R20, HIGH(RAMEND) ; MAIN ROUTINE
                OUT SPH, R20
                LDI R20, LOW(RAMEND)
                OUT SPL, R20; RELOCATE STACK POINTER

                SBI DDRB, 1; PB1 - OUTPUT
                SBI DDRB, 7; PB7 - OUTPUT


      ; ------- TIMER 0 AND TIMER 2 INTERRUPT ENABLING USING THE ATMEGA32 AND ATMEGA324PB -
;               LDI R20, (1 << TOIE0) | (1 << TOIE2)
;               OUT TIMSK, R20; ENABLE TIMER 0 AND TIMER 2 OVERFLOW INTERRUPT SOURCES - THIS IS THE REGISTER FOR THE ATMEGA32
```

```
		; ------ ATMEGA324PB INTERRUPT ENABLING
		LDI R20, 0x01; 0000 0001 - TOIE0
		; TIMSK0 IS AT I/O LOCATION 0x6E SO CANNOT USE OUT TIMSK0, R20 BECAUSE OUT IS LIMITED TO 0-63 (DECIMAL)
		STS 0x6E, R20; USE THIS INSTEAD

		LDI R20, 0x01; 0000 0001 - TOIE2
		; TIMSK2 IS AT I/O LOCATION 0x70 SO CANNOT USE OUT TIMSK2, R20 BECAUSE OUT IS LIMITED TO 0-63 (DECIMAL)
		STS 0x70, R20; USE STS INSTEAD
		; ---------------------- END TIMER 0 AND TIMER 2 INTERRUPT ENABLING




		; ----- TIMER 0 CONFIGURATION FOR NORMAL, INTERNAL CLOCK, NO PRESCALER FOR ATMEGA32 AND ATMEGA324PB
		; --- TCNT0 IS THE SAME FOR ATMEGA32 AND FOR ATMEGA324PB
		LDI R20, -160 ;
		OUT TCNT0, R20; INITIALIZATION FOR 10 us - fOSC = 16 MHz  => 0.0625 us/MC  => 0.0625 us/MC * 160 = 10 US


		; ----- ATMEGA32
;		LDI R20, 0x01; TIMER 0 NORMAL, INTERNAL CLOCK, NO PRESCALER
;		OUT TCCRO, R20

		; ----- ATMEGA324PB
		LDI R20, 0x00
		OUT TCCR0A, R20; NORMAL MODE
		LDI R20, 0x01; INTERNAL CLK, NO PRESCALER
		OUT TCCR0B, R20;
		; ------ END TIMER 0 CONFIGURATION

		; ------TIMER 2 CONFIGURATION FOR NORMAL, INTERNAL CLOCK, NO PRESCALER FOR ATMEGA32 AND ATMEGA324PB
		; TCNT2 IS THE SAME FOR BOTH ATMEGA32 AND ATMEGA324PB
		LDI R20, -80; VALUE FOR 5 us
		; TCNT2 IS AT 0xB2 WHICH IS ABOVE 0-63 SO CANNOT USE OUT TCNT2, R20   for ATmega324pb
		STS 0xB2, R20; USE THIS INSTEAD   for ATmega324pb
		;OUT TCNT2, R20; for ATmega32
		; ----- ATMEGA32
;		LDI R20, 0x01
;		OUT TCCR2, R20; TIMER 2 - NORMAL MODE, INTERNAL CLOCK, NO PRESCALER

		; ----- ATMEGA324PB
		LDI R20, 0x00
```

```
; TCCR2A IS AT 0xB0 SO CANNOT USE OUT TCCR2A, R20; NORMAL MODE
STS 0xB0, R20; USE THIS INSTEAD

LDI R20, 0x01
; TCCR2B IS AT 0xB1 SO CANNOT USE OUT TCCR2B, R20; INTERNAL CLK, NO PRESCALER
STS 0xB1, R20; USE THIS INSTEAD
;-----END TIMER 2 CONFIGURATION

LDI R20, 0x00
OUT DDRC, R20; PORT C - INPUT PORT

LDI R20, 0xFF
OUT DDRD, R20; PORT D- OUTPUT PORT


SEI; SET I = 1 IN SREG - GLOGAL INTERRUPT ENABLE

;---------- MAIN LOOP
HERE:   IN R20, PINC; READ FROM PORT C
        OUT PORTD, R20; COPY TO PORT D
        JMP HERE
;---------- END MAIN LOOP

;---------- TIMER 0 ISR
.ORG 0x200
T0_OV_ISR:  LDI R16, -40; -160; 10 us
            OUT TCNT0, R16; LOAD TIMER 0 FOR NEXT ROUND FOR 10 us
HALF WAVE
            IN R16, PORTB; NOT PINS BECAUSE USING PORTB.1, PORTB.7 AS
OUTPUTS
            LDI R17, 0x02; 0000 0010 - TOGGLE PB1 FOR OUTPUT
WAVEFORM FOR TIMER 0
            EOR R16, R17; SINCE XORING PB7 WITH 0 - PB7 IS UNCHANGED
            OUT PORTB, R16; TOGGLE PB1
            RETI; RETURN FROM TIMER 0 OVERFLOW ISR

;---------- TIMER 2 ISR
.ORG 0x300
T2_OV_ISR:      LDI R18, -30; -80; 10 us
            ; TCNT2 IS AT 0xB2 SO CANNOT USE OUT TCNT2, R18; LOAD
TIMER 2 FOR NEXT ROUND FOR 5 us HALF WAVE
            STS 0xB2, R18; USE THIS INSTEAD
            IN R18, PORTB; NOT PINS BECAUSE USING PORTB.1, PORTB.7 AS
OUTPUTS
            LDI R19, 0x80; 1000 0000 - TOGGLE PB7 FOR OUTPUT
WAVEFORM FOR TIMER 0
            EOR R18, R19; SINCE XORING PB1 WITH 0 - PB1 IS UNCHANGED
            OUT PORTB, R18; TOGGLE PB7
```

RETI; RETURN FROM TIMER 2 OVERFLOW ISR

ATMEGA324PB I/O REGISTERS
EIFR P. 81 (EXTERNAL INTERRUPT FLAGS)
TIMSK0 P. 130 (TIMER0 ONLY) TIMSK2 P. 201 (TIMER2 ONLY)
TCCR0A AND TCCR0B FOR TIMER0 (2 OUTPUT COMPARE A AND B) (STARTS P. 126)
TCCR2A AND TCCR2B FOR TIMER2 (P. 196)
TIFR0 P. 133 (TIMER0 ONLY), TIFR2 P. 202 (TIMER2 ONLY)
EIMSK P. 80 (EXTERNAL INTERRUPT ENABLE
PINOUT FOR ATMEGA324PB P. 14
INTERRUPT VECTOR TABLE ATMEGA324PB P. 73

```
/*
 * TimerInterruptCExampleAVR.c
 *
 * Author : stanleyj
 */

#include <avr/io.h>
#include <avr/interrupt.h>

ISR (TIMER0_OVF_vect);
ISR (TIMER2_OVF_vect);

int main(void)
{
	DDRB = 0x82;  // MAKE DDRB.1 AND DDRB.7 OUTPUT
	DDRC = 0x00;   // PORTC INPUT
	PORTC = 0xFF;  // ENABLE PULL-UP RESISTORS FOR INPUT PORT C
	DDRD = 0xFF;   // MAKE PORTD OUTPUT
		-160;
	TCNT0 = [illegible];   // [illegible] TIMER 0 PRELOAD INITIALIZATION FOR HALF WAVE
//	TCCR0 = 0x01;  // TIMER 0 NORMAL MODE, CLK, NO PRESCALER - ATMEGA32
	TCCR0A = 0x00; // TIMER 0 NORMAL MODE - ATMEGA324PB
	TCCR0B = 0x01; // TIMER 0 CLK, NO PRESCALER - ATMEGA324PB
		-80;
	TCNT2 = [illegible];   // [illegible] TIMER 2 PRELOAD INITIALIZATION FOR HALF WAVE
 // TCCR2 = 0x01;  // TIMER2 NORMAL MODE, CLK, NO PRESCALER - ATMEGA32
	TCCR2A = 0x00; // TIMER 2 NORMAL MODE - ATMEGA324PB
	TCCR2B = 0x01; // TIMER 2 CLK, NO PRESCALER - ATMEGA324PB

//	TIMSK = (1<<TOIE0) | (1<<TOIE2); // ENABLE TIMER 0 AND TIMER 2 INTERRUPTS - ATMEGA32
	TIMSK0 = 0x01;       // TIMER 0 OVERFLOW INTERRUPT ENABLE
	TIMSK2 = 0x01;  // TIMER 2 OVERFLOW INTERRUPT ENABLE

	sei();          // I = 1 IN SREG FOR GLOBAL INTERRUPT ENABLE

	while (1)
	{
		PORTD = PINC; // ENDLESS LOOP - READ PORTC PINS AND COPY TO PORTD
	}


}


// ----- TIMER 0 OVERFLOW ISR
ISR (TIMER0_OVF_vect)
{
		-160;
	TCNT0 = [illegible];    // [illegible] RESET PRELOAD FOR TIMER 0 FOR HALF WAVE
	PORTB ^= 0x02;  // TOGGLE PB1 FOR TIMER 0 WAVE FORM
}


// ------ TIMER 2 OVERFLOW ISR
ISR (TIMER2_OVF_vect)
{
		-80;
	TCNT2 = [illegible];    // [illegible] RESET PROLOAD FOR TIMER 2 FOR  HALF WAVE
	PORTB ^= 0x80;
}
```

latency - time between when an interrupt source is activated & when the source is serviced

4 cycles { push PC onto stack
load PC with ISR vector address

MUL 4 MCs
RET

LDI 1 MC

